LEVEL 4 · PART II

Docker

4

Goal: Go beyond basic container usage and understand how Docker works internally.

RECOMMENDED TIME	TOPICS
2-3 weeks	14

PRACTICE / CAPSTONE

Containerize a Node.js API, PostgreSQL, Redis, Nginx, Prometheus, Grafana and Loki using Docker Compose.

Topics

#	Topic	What you will learn
1	How Docker Works Internally **p.739**	namespaces, cgroups, union filesystems, containerd/runc, Docker vs VMs
2	Images and Layers **p.751**	image anatomy, content-addressable layers, tags vs digests
3	Containers and Their Lifecycle **p.761**	run/stop/exec/logs, PID 1, signals, restart policies
4	Registries **p.773**	Docker Hub, ECR, GHCR, tagging/pushing, private registries
5	Volumes and Bind Mounts **p.785**	named volumes, bind mounts, tmpfs, persistence and backups
6	Docker Networks **p.795**	bridge/host/none/overlay, DNS between containers, port publishing
7	Dockerfile **p.805**	instructions, CMD vs ENTRYPOINT, .dockerignore, ARG vs ENV
8	Docker Compose **p.817**	services, networks, volumes, depends_on with healthchecks, profiles, env files
9	Image Caching and Layer Optimization **p.829**	cache invalidation order, BuildKit cache mounts, size reduction
10	Multi-stage Builds **p.841**	build vs runtime stage, copying artifacts, target stages
11	Non-root Containers **p.851**	USER, file ownership, dropping capabilities, read-only rootfs
12	Health Checks **p.865**	HEALTHCHECK, liveness vs readiness concepts, Compose health gating

#	Topic	What you will learn
13	Minimal Base Images **p.877**	alpine vs slim vs distroless vs scratch, glibc/musl trade-offs
14	Image Scanning and Container Security **p.889**	Trivy/Grype, CVEs, SBOM, signing basics, runtime hardening
★	Level capstone project **p.903**	Containerize a Node.js API, PostgreSQL, Redis, Nginx, Prometheus, Grafana and Loki using Docker Compose.

How to work through this level

1. Read each topic in order: *what it is* → *why it matters* → *real-world example* → *diagram* → *guide*.
2. Do **every task** in the topic's *Tasks* section and tick the box (`- [x]`) in your own fork or notes.
3. Answer the *Self-check questions* without looking, then reveal the answers.
4. Finish with the **capstone project**, which combines all topics of the level.
5. Move on only when you can do the capstone's *Prove it* task without the guide.

Topic flow

